DATA SCIENCE & ARTIFICIAL INTELLIGENCE

Probability and Statistics

Statistics - 2

only for DS/AI

Lecture No. 01



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Recap of previous lecture



Topic

CORRELATION - REGRESSION

Topics to be Covered



Topic

JOINT PROBABILITY MASS FUNCTION (joint p.m.f)

(DISCRETE CASE)

Joint P.M.F / Joint Prob Dist.

① If X & Y are two Random Variables then joint p.m.f is given as

$$p_{ij} = P[x_i, y_j] = P[X = x_i \ \& \ Y = y_j]$$

$$p_{ij} \geq 0$$

$$\sum_{i=1}^{m} \sum_{j=1}^{n} p_{ij} = 1$$

② Individual Prob Dist. of X & Y are called M.D.f defined as,

X :	x_1	x_2	x_3	----	x_m
$P(X)$:	p_1	p_2	p_3	---	p_m

&

Y :	y_1	y_2	y_3	----	y_n
$P(Y)$:	p_1'	p_2'	p_3'	---	p_n'

③ X & Y are called Ind R.V if $\forall$ x_i & y_j we have $P(X = x_i, Y = y_j) = P(X = x_i) \cdot P(Y = y_j)$

Conditional Dist for joint p.m.f is given as;

$$P\left[\frac{X=x_i}{Y=y_j}\right]=\frac{P(x_i,y_j)}{P(Y=y_j)} \quad \& \quad P\left[\frac{Y=y_j}{X=x_i}\right]=\frac{P(x_i,y_j)}{P(X=x_i)}$$

[NAT]

#Q. A fair ~~aim~~ coin is tossed thrice. If x denote the number heads on the 1st toss and Y denotes the total number of heads then write it's joint p.m.f.

(ii) Also find marginal distributions of x and y

(iii) Evaluate $P(x = 1, y \geq 1)$, $P(x \geq 0, y \geq 2)$, $P(0 \leq x \leq 1, y = 3)$

(iv) Conditional distribution of x when $y = 1$

(v) Conditional distribution of ~~x~~ y when $x = 1$

(vi) $P(x = y)$,

(vii) $P(1 \leq y \leq 3) = ?$

(viii) $P(y = 3) = ?$

Sol: $S = \{(HHH), (HHT), (HTH), (HTT), (THH), (THT), (TTH), (TTT)\} = 8$

$X = \{\text{No. of Heads on } 1^{st} \text{ toss}\} = \{0, 1\}$, $Y = \{\text{Total no. of Heads}\} = \{0, 1, 2, 3\}$

X:	0	1
$P(X)$:	$\frac{4}{8}$	$\frac{4}{8}$

= Marginal Dist of X

Y:	0	1	2	3
$P(Y)$:	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

Marginal Dist of Y

X \ Y	0	1	2	3	$\Sigma(X)$
0	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{1}{8}$	0	$\frac{4}{8}$
1	0	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{1}{8}$	$\frac{4}{8}$
$\Sigma(Y)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$	$= 1$

* This Table is Called joint P.M.F for X & Y
* Individual Prob Dist of X & Y are Called Marginal Distributions

(iii) $P(X=1, Y\geq 1) = ? = P(X=1 \& Y=1,2,3)$

$= P[(1,1) \text{ or } (1,2) \text{ or } (1,3)]$

$= \frac{1}{8} + \frac{2}{8} + \frac{1}{8} = \frac{4}{8} = \frac{1}{2}$

It is Prob, when 1st toss is showing Head and total no of Heads are at least one.

$P(X\geq 0, Y\geq 2) = ? = P(X=0 \text{ or } 1 \& Y=2 \text{ or } 3)$

$= P[(0,2) \text{ or } (0,3) \text{ or } (1,2) \text{ or } (1,3)]$

$= \frac{1}{8} + \frac{0}{8} + \frac{2}{8} + \frac{1}{8} = \frac{4}{8} = \frac{1}{2}$

$P(0\leq X\leq 1, Y=3) = ?$

$= P(X=0 \text{ or } 1 \& Y=3)$

$= P[(0,3) + (1,3)] = \frac{0}{8} + \frac{1}{8} = \frac{1}{8}$

(vi) $P(X=Y) = ? = P[(0,0) \text{ or } (1,1)]$

$= \frac{1}{8} + \frac{1}{8} = \frac{1}{4}$

(viii) $P(Y=3) = ? = \frac{1}{8}$ (By M.D.F of Y)

(M-II) $P(Y=3) = P(X=0 \text{ or } 1 \& Y=3)$

$= P[(0\,3) \text{ or } (1\,3)] = \frac{0}{8} + \frac{1}{8} = \frac{1}{8}$

(vii) : $P(1 \le Y \le 3) = ?$

(M-I)

Y:	0	1	2	3
$P(Y)$:	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

$$\text{So } P(1 \le Y \le 3) = P(Y = 1 \text{ or } 2 \text{ or } 3)$$

$$= \frac{3}{8} + \frac{3}{8} + \frac{1}{8} = \frac{7}{8}$$

(M-II)

$$P(1 \le Y \le 3) = P(X = 0 \text{ or } 1 \text{ \& } 1 \le Y \le 3)$$

$$= P(X = 0 \text{ or } 1 \text{ \& } Y = 1 \text{ or } 2 \text{ or } 3)$$

$$= P[(01) + (02) + (03) + (11) + (12) + (13)]$$

$$= \frac{2}{8} + \frac{1}{8} + \frac{0}{8} + \frac{1}{8} + \frac{2}{8} + \frac{1}{8} = \frac{7}{8}$$

(iv) (Conditional dist of X when $Y=1$) $\rightarrow P(X=x_i / Y=1) = \frac{P[(x_i, 1)]}{P(Y=1)}$

$$P(X=0/Y=1) = \frac{P(0,1)}{P(Y=1)} = \frac{2/8}{3/8} = \frac{2}{3}$$

$$P(X=1/Y=1) = \frac{P(1,1)}{P(Y=1)} = \frac{1/8}{3/8} = \frac{1}{3}$$

For req. Cond^{n} Distn

X :	0	1
$P(X=x/Y=1)$:	$2/3$	$1/3$

(V): (Conditional Dist. of Y, when $x=1$) $\rightarrow P\left[\frac{Y=y_j}{x=1}\right] = \frac{P(1, y_j)}{P(x=1)}$

$$P(Y=0/x=1) = \frac{P(1,0)}{P(x=1)} = \frac{0}{4/8} = 0$$

$$P(Y=1/x=1) = \frac{P(1,1)}{P(x=1)} = \frac{1/8}{4/8} = \frac{1}{4}$$

$$P(Y=2/x=1) = \frac{P(1,2)}{P(x=1)} = \frac{2/8}{4/8} = \frac{2}{4}$$

$$P(Y=3/x=1) = \frac{P(1,3)}{P(x=1)} = \frac{1/8}{4/8} = \frac{1}{4}$$

Req Cond^{n} Dist^{n} is

Y :	0	1	2	3
$P(Y=y_j/x=1)$:	0	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

[NAT]

#Q. Four coins are tossed, let x denotes numbers of heads and y denotes (number of Head-number of tails). Then write the joints pmf for such type of random experiment.

(i) Evaluate $P(-2 \le y < 4)$, (ii) $P(x \le 2, y \ge 2)$

(iii) $P(x \ge 3, -2 < y \le 4)$,

(iv) Conditional distribution of x given the value of y = 0

(v) Conditional distribution of y given the value of x = 3

(vi) Marginal distribution of x

(vii) Marginal distribution of y

(viii) $P(x = y) = ?$ (Ix) $p(x \le 2) = ?$

$S = \{(HHHH), (HHHT), (HHTT), (HTTT), (TTTT)\} \Rightarrow n(S) = 2^4 = 16$ Quadruples.
$\quad\quad\quad

joint Dist Table for X & Y →

X \ Y	-4	-2	0	2	4	Σ(X)
0	1/16	0	0	0	0	1/16
1	0	4/16	0	0	0	4/16
2	0	0	6/16	0	0	6/16
3	0	0	0	4/16	0	4/16
4	0	0	0	0	1/16	1/16
Σ(Y)	1/16	4/16	6/16	4/16	1/16	1

∴ Her Diff may be −4 or −2 or 0

(ii) $P(X \le 2, Y \ge 2) = ?$

$= P(X = 0 \text{ or } 1 \text{ or } 2 \;\&\; Y = 2 \text{ or } 4)$

$= P[(0,2) \text{ or } (0,4) \text{ or } (1,2) \text{ or } (1,4) \text{ or } (2,2) \text{ or } (2,4)]$

$= 0+0+0+0+0+0 = 0$

it is an Impossible Event because we will never see that No. of Heads are 0 or 1 or 2 & Diff b/n H & T is 2 or 4

(iii) $P(X \geq 3, -2 < Y \leq 4) = ?$

$= P(X = 3 \text{ or } 4 \;\&\; Y = 0 \text{ or } 2 \text{ or } 4)$

$= P[(3\,0) \text{ or } (3\,2) \text{ or } (3\,4) \text{ or } (4\,0) \text{ or } (4\,2) \text{ or } (4\,4)]$

$= 0 + \frac{4}{16} + 0 + 0 + 0 + \frac{1}{16} = \frac{5}{16}$

(viii) $P(X = Y) = ?$

$= P[(0\,0) \text{ or } (1\,1) \text{ or } (2\,2) \text{ or } (3\,3) \text{ or } (4\,4)]$

$= 0 + 0 + \frac{1}{16} = \frac{1}{16}$

(iv) (Cond^{n} Dist of X for given $Y=0$)

X	0	1	2	3	4
$P(X=x_i / Y=0)$	0	0	1	0	0

$$P(X=2/Y=0) = \frac{P(2,0)}{P(Y=0)} = \frac{6/16}{6/16} = 1$$

(v) (Cond^{n} Dist of Y for $X=3$)

Y	-4	-2	0	2	4
$P(Y=y_j / X=3)$	0	0	0	1	0

$$P(Y=2/X=3) = \frac{P(3,2)}{P(X=3)} = \frac{4/16}{4/16} = 1$$

[NAT]

#Q. Joint P.M.F. for random variables x and y are given in the table. Find $P[1 \le x < 3, y \ge 1] = ?$

x \ y	0	1	2	p(x)
0	$\frac{1}{18}$	$\frac{1}{9} = \frac{2}{18}$	$\frac{1}{6} = \frac{3}{18}$	1/3
1	$\frac{1}{9} = \frac{2}{18}$	$\frac{1}{18}$	$\frac{1}{9} = \frac{2}{18}$	5/18
2	$\frac{1}{6} = \frac{3}{18}$	$\frac{1}{6} = \frac{3}{18}$	$\frac{1}{18}$	7/18
p(y)	1/3	1/3	1/3	

$P[1 \le x < 3, y \ge 1] = ?$ $= P[x = 1 \text{ or } 2 \;\&\; y = 1 \text{ or } 2]$

$= P[(11) \text{ or } (12) \text{ or } (21) \text{ or } (22)]$

$= \frac{1}{18} + \frac{1}{9} + + \frac{1}{6} + \frac{1}{18}$

$= \frac{1}{18} + \frac{2}{18} + \frac{3}{18} + \frac{1}{18} = \frac{7}{18}$ Ans

(ii) M.Df of X is

X:	0	1	2
$P(x)$:	$1/3$	$\frac{5}{18}$	$\frac{7}{18}$

& MDf of Y is

Y:	0	1	2
$P(y)$:	$1/3$	$1/3$	$1/3$

[NAT]

#Q. Joint p.d.f of two discrete Random variables x and y is given as $f(x, y) = c(2x + y)$ where $0 \le x \le 2$ and $0 \le y \le 3$ then

(i) Find C

(ii) $P[(x = 2), y = 1]$

(iii) $P(x \ge 1, y \le 2) = ?$

AT Concept $\sum\sum p_{ij} = 1$

① $42c = 1 \Rightarrow c = \frac{1}{42}$

x \ y	0	1	2	3	P(x)
0	0	c	2c	3c	
1	2c	3c	4c	5c	
2	4c	5c	6c	7c	
P(y)					1

(ii) $P(X=2, Y=1) = ? = P(2,1) = 5C = \frac{5}{42}$

(iii) $P(X \geq 1, Y \leq 2) = ? = P[X = 1 \text{ or } 2 \;\&\; Y = 0 \text{ or } 1 \text{ or } 2]$

$= P[(10) \text{ or } (11) \text{ or } (12) \text{ or } (20) \text{ or } (21) \text{ or } (22)]$

$= 2C + 3C + 4C + 4C + 5C + 6C$

$= 24C = 24\left(\frac{1}{42}\right) = \frac{12}{21} = \frac{4}{7}$

[NAT]

#Q. X and Y are two random variable having the joint density function,

$$f(x, y) = \frac{1}{27}(2x + y)$$

Where x and y can only be integer 0, 1 or 2. Find

(i) Marginal distributions of x and y

(ii) Conditional distribution of y given x = 1

x \ y	0	1	2	P(x)
0	0	1/27	2/27	3/27
1	2/27	3/27	4/27	9/27
2	4/27	5/27	6/27	15/27
P(y)	6/27	9/27	12/27	= 1

y	0	1	2
$P(Y=y_j/x=1)$	2/9	1/3	4/9

[NAT]

HW8

#Q. Two Dice are rolled simultaneously. Let x denotes the sum of two faces and y the absolute value of their difference. Find joint p.m.f for x and y.

(i) Also evaluate P (x = y) Ans = 0

(ii) P [x = 7 ∩ y ≥ 3] Ans = 1/9

good

(iii) P [(x + y = 4) ∩ (x − y = 2)] = ? Ans = 1/18

(iv) Find marginal distributions of x and y

(v) Find conditional distribution of x given y = 2

$X = \{2, 3, 4, 5, \ldots, 11, 12\}$

$Y = \{0, 1, 2, 3, 4, 5\}$

THANK - YOU